

Project Demo Planning

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Context Setup	Introduce the OptiCrop platform, articulate the primary customer problem statements (Farmers, Researchers, Policymakers), and outline the project objectives.	2 mins	Hari Charan
2	Technical & Architectural Overview	Walk through the technology stack choices (Flask, Scikit-Learn, Bootstrap 5) and explain how data moves from the input form to the deserialized model.pkl binary matrix.	2 mins	Hari Charan
3	Model Training & Compilation Loop	Demonstrate the terminal-based execution of train_model.py, showcasing the explicit multi-model evaluation loop and printing model accuracy ranks.	2 mins	Hari Charan
4	Live Feature Demo: Scenario 1 & 2	Boot up the Flask application and show live form submissions for Scenario 1 (Predictive Crop Recommendations) and Scenario 2 (Target Crop Suitability Auditing / Mismatch Tracking).	3 mins	Hari Charan
5	Live Feature Demo: Scenario 3	Input custom parameters into Scenario 3 to demonstrate how the engine renders dynamic strategic regional resource optimization guidelines for policymakers.	2 mins	Hari Charan
6	Performance & Code Quality Summary	Present the performance test summary metrics (sub-400ms average inference speeds) and the clean, modular code properties built to bypass Pylance diagnostic warnings.	1 mins	Hari Charan

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Focus heavily on how climate shifts alter traditional agricultural habits, necessitating real-time mathematical validation over multi-variable parameters (N, P, K, pH, etc.).
2	Solution Overview	Emphasize that your system runs self-contained locally via Anaconda distribution package configurations, making deployment fast and avoiding compilation failures.
3	Live Feature Demonstration	Key Highlight: Purposely input an out-of-bounds parameter value during the walkthrough to show how the backend's explicit validation layer cleanly catches errors instead of crashing the system loop.
4	Q&A Session	Be prepared to discuss the optimization choice of the Random Forest algorithm compared to KNN and Logistic Regression variants based on your terminal logs.